HCSA Rev D blueprint pack: the FEA-sized baseline

Source: HCSA Rev D design spec (14 Oct 2026), the only source of numbers on these sheets. Shell geometry computed from the truncated-icosahedron vertex construction (hcsa_geom.py). Anything the spec does not give is left undimensioned and listed as 'not specified'.

Index

RD-00 Cover, index, change table, conventions (this sheet)
RD-01 Shell: elevation and plan, panel names, vertex and seam counts
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RD-04 Rib sections: main hex, main pent, secondary (and Rev S superseded)
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RD-12 Open decisions and what each test settles

Colour and convention key

Frame A (flat joint face): light blue. Frame B (grooved face): tan. Steel / A286: grey-gold.
Glass (L1, L7): light cyan. O-ring: red. RTV566: yellow. Centrelines: blue dash-dot. Hidden: dashed.
Superseded Rev S geometry: grey, crossed. PROPOSAL stamp (magenta): sized in the 16 Oct submodels.
Dimensions in mm, no scale: do not scale the sheets. Section letters A-A, B-B on the sheet that cuts them.
LOCKED / FEA / PROPOSAL / OPEN as in the spec status tags.

Coordinate convention

z: depth from the space (outer) face inward, z 0 at the outer face of the frame flange; z 140 is the frame's cabin face, z 375 the main-rib cabin flange. Ribs are depth into the cabin from the outer face.
x: along a seam (0 at a vertex, 1,125 at the seam mid-point, 2,250 at the next vertex) or along a rib (0 at the vertex, 2,250 at the centroid).
y: across the joint plane, 0 on the joint plane, positive into frame A.
Joint planes are radial (contain the seam and the shell centre); the miter is 20.905 deg on a hexagon, 16.472 deg on a pentagon.

What changed	Rev S	Rev D	Why (the FEA result)
Hexagon glazing	6 panes, 2.19 m2 each, 120 on the shell	24 triangular panes, 1,125 side, 0.548 m2, 480 on the shell	Rev S L7: 54 MPa vs 35 allowable, 7.3 mm vs 3.1 (whole-shell runs, 10 Oct)
L7 pressure pane	31 mm BOROFLOAT 33	25.4 mm BOROFLOAT 33 (thickest standard)	24-pane L7: 22.4 to 22.9 MPa (27 mesh-corrected) vs 35; bending 1.03 mm
Hexagon main rib	I 103, flanges 64 × 10, web 12	I 375, flanges 200 × 15, web 8 (A 8,760, I 2.22e8)	Rev S ribs about 1,470 MPa; Rev D 106 to 128 MPa vs 160 allowable
Pentagon main rib	I 103 (as hexagon)	I 400, flanges 80 × 25, web 8 (A 6,800)	114 to 140 MPa vs 160
Secondary ribs	none	18 per hexagon, I 160, 64 × 8, web 6; 1,125 long	116 to 147 MPa with the hardpoint pinned (I 200, 100 × 10 if moment-stiff: OPEN)
Corner	frames meet on the miter, open	machined solid corner block, first 100 mm of both frames; 4 × 5/8-18 A286 vertex ties per joint face, 2 dowels dia 12	Rev S corner 140 to 480x too soft in translation, 4 to 9x in rotation; end bolts 100 to 138 kN vs 34 kN break
Seam mid hardpoint	none	lug 40 / clevis 2 × 25 / pin dia 30 A286 at x 1,125, block 150 long	31 % of the seam tension (170 to 195 kN) passes here
Seam bolt role and count	carries the seam tension, 20.7 kN per bolt; 29 per seam	seal clamp, 2 to 12 kN in service (10 kN preload); 29 stations at 75 pitch (x 75 to 2,175), stations 1 and 29 replaced by the vertex ties and station 15 by the hardpoint pin: 26 bolts per seam, 2,340 on the shell (PROPOSAL, C1 / C3)	D1: Rev S slice pries open at 0.3 atm, passes 17 kN per bolt at 0.66 atm; the FEA ran 29 per seam, the difference is inside the 5 % the bolts carry
Seam bolt length	1-3/4 in	2-1/4 in (57) minimum, grip 43.2	1-3/4 was short for two 20 webs and two 1.6 washers (fixed 14 Oct)
O-ring gland	6.6 × 4.0	7.1 × 4.0 (PROPOSAL)	Parker face-seal practice for a 5.33 cord: fill 78 %, squeeze 25 %
Seal cord	FVMQ	butyl BA500 closed loop (default), FVMQ alternative: OPEN	Parker, 5 Oct: fluorosilicone permeates in vacuum; butyl is the only cord to -40 C
Pentagon plate	4 mm	10 mm 6061-T6	about 120 MPa mesh-corrected
L7 setting strip	2 mm	2 mm; 1.0 mm 80 Shore A alternative: OPEN	seat movement 1.6 to 1.8 mm against the 1.62 limit (1/200 of 325)
Frame	140 deep	140 deep kept; bow 1.4 to 2.0 mm vs 1 mm limit: OPEN	a 300 mm frame still bows 1.07; test (T1, C2) before deepening
Mass	structure 25.4 t, facility 41.7 t	structure 33.3 t, facility about 50.9 t; 25 t target withdrawn	Rev S fails; Falcon Heavy reusable no longer a single-launch option

Status

Every sheet carries the Rev D footer. PROPOSAL items (corner node, vertex ties, hardpoint, 7.1 gland) are drawn to carry the FEA loads and are sized on 16 Oct (C1, C3, C4). Items with no value in the spec are drawn but not dimensioned and listed as 'not specified' on the sheet. Section properties on RD-04 are computed from the section dimensions (shown as calc.); stresses and allowables are the spec's.

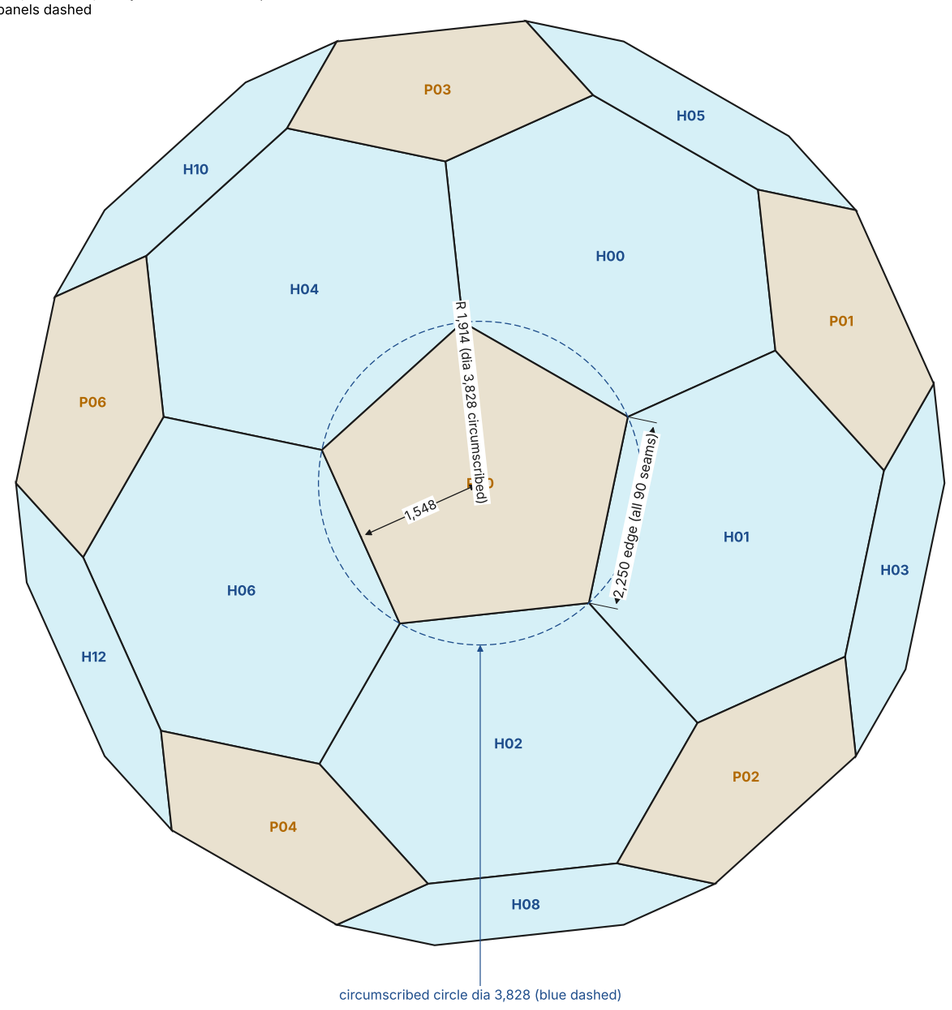
Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision.

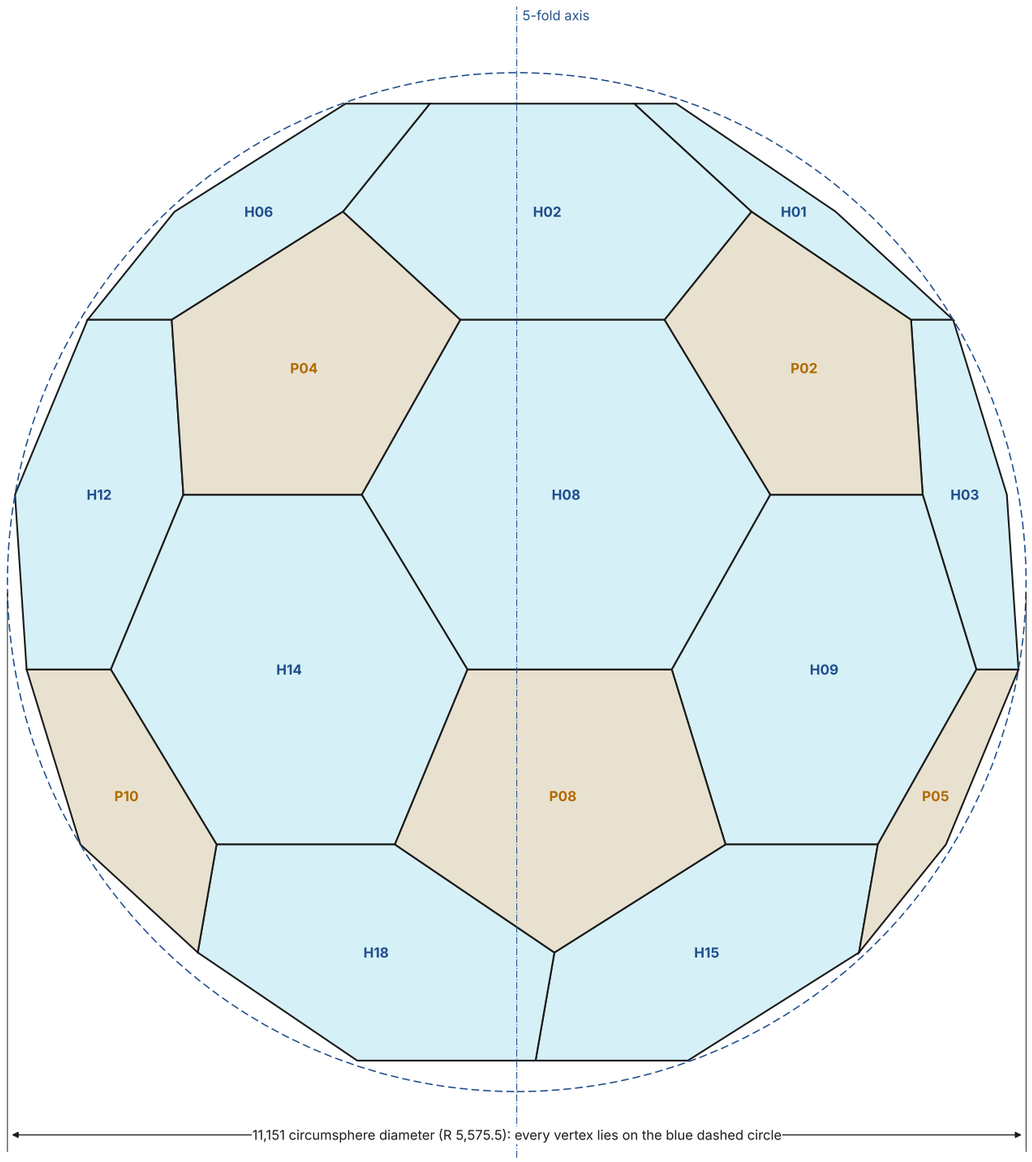
HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 00: cover, index, Rev S to Rev D changes, conventions	**DWG RD-00**
Material: n/a (no part on this sheet)	
HT: n/a	

Plan, looking down the 5-fold axis onto pentagon P00 (space side)

H: 20 hexagons (cyan) P: 12 pentagons (tan)
60 vertices, 90 seams; every vertex is hex-hex-pent
hidden panels dashed



Elevation, 5-fold axis vertical, viewed along y



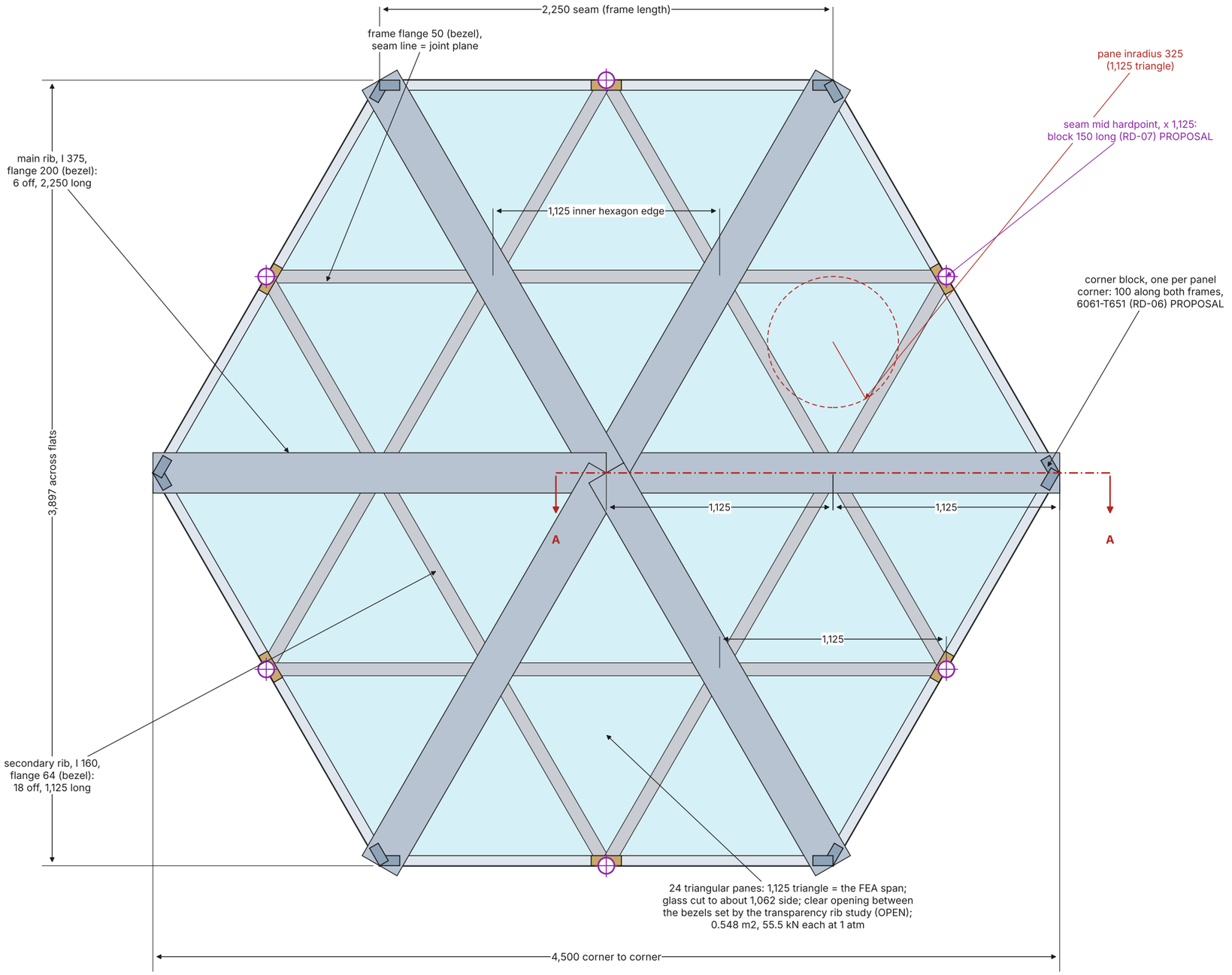
Shell numbers (spec section 1)
Topology: truncated icosahedron, 20 hexagons, 12 pentagons, 60 vertices, 90 seams. Edge 2,250. Circumradius 5,575.5 (dia 11,151). Surface 367.574 m2.
Hexagon: 4,500 corner to corner, 3,897 across flats, 13.153 m2 (RD-02). Pentagon: 1,914 circumradius, 1,548 inradius, 8.710 m2 (RD-09).
Dihedral: hex-hex 138.19 deg, hex-pent 142.62 deg. Joint planes radial; miter hex 20.905 deg, pent 16.472 deg.
Pressure 101.325 kPa: hexagon face load 1,333 kN, pentagon 883 kN; seam tension 276.7 kN/m (623 kN per seam).
Panel names H00..H19, P00..P11 follow the geometry script's face order (hcsa_geom.py build()).

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision.

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 01: shell, elevation and plan, panel names	**DWG RD-01**
Material: 6061-T651 plate / 6061-T6511 extrusion (prototype); 2219-T851 (flight); L7 BOROFLOAT 33	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C	

Hexagon panel, viewed from the cabin (z 375 flange side); one of 20



Notes (spec section 2)
6 main ribs, centroid to each vertex, 2,250 long, split at the mid-point (x 1,125 from the vertex).
18 secondary ribs, all 1,125 long: 6 join the main-rib mid-points into an inner hexagon of 1,125 edge; 12 join each main-rib mid-point to the two neighbouring seam mid-points, so two ribs of each panel land at every seam mid-point (four per hardpoint).
24 equilateral panes, 1,125 side (the FEA span), inradius 325, 0.548 m2 each; glass cut to about 1,062; clear opening between the 200 / 64 / 50 bezels set by the transparency rib study (OPEN).
Aluminium per hexagon: frame 13.5 m, main ribs 13.5 m, secondary ribs 20.25 m.
Face load at 1 atm 1,333 kN; 480 panes on the shell.
Rev S (6 panes of 2.19 m2, 31 mm glass) superseded: 54 MPa vs 35 and 7.3 mm vs 3.1.

Bezels drawn at the flange widths: main 200, secondary 64, frame 50 from the seam. Rib depths on RD-03 / RD-04.
Corner blocks and hardpoint blocks: PROPOSAL (RD-06, RD-07).
Seam bolts: 26 per seam at the 75 pitch stations not taken by the corner blocks and the hardpoint (RD-05) PROPOSAL.
Not specified: pane edge clearance at ribs (frame value 10 clear, 20 bite on RD-08), rib splice at the mid-point, rib-to-rib joint.

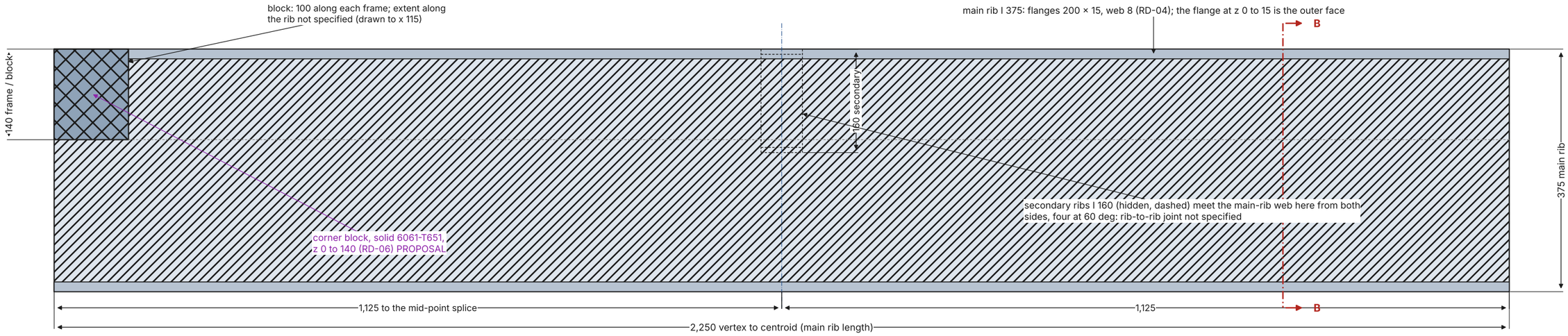
PROPOSAL items: corner blocks, hardpoints

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision. Rib-to-rib and rib-to-block joints: not specified (C1).

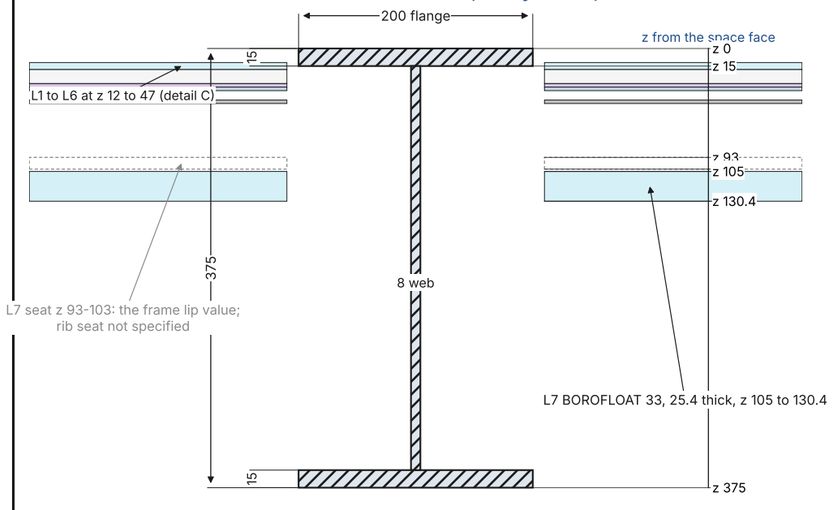
HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 02: hexagon panel plan, cabin side	**DWG RD-02**
Material: 6061-T651 plate / 6061-T6511 extrusion (prototype); 2219-T851 (flight)	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C	

Section A-A (RD-02): x along the main rib, 0 at the vertex, 2,250 at the centroid; z down from the space face

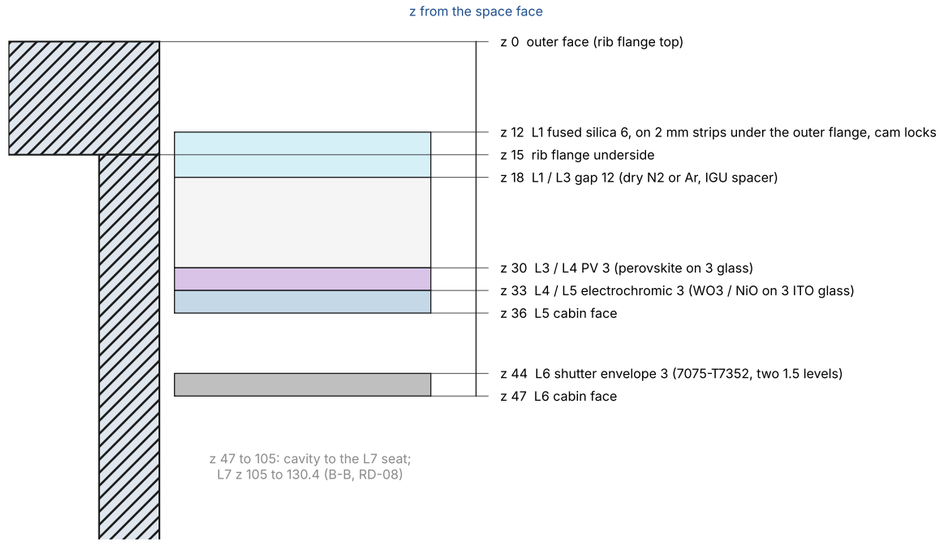


The glazing lies beside the rib and is hidden by the cut web in this plane: stack z-ordinates in B-B and detail C below.

Section B-B across the main rib at x 1,900 (y across, z down)



Detail C: L1 to L6 z-ordinates beside the rib (enlarged); L7 in B-B



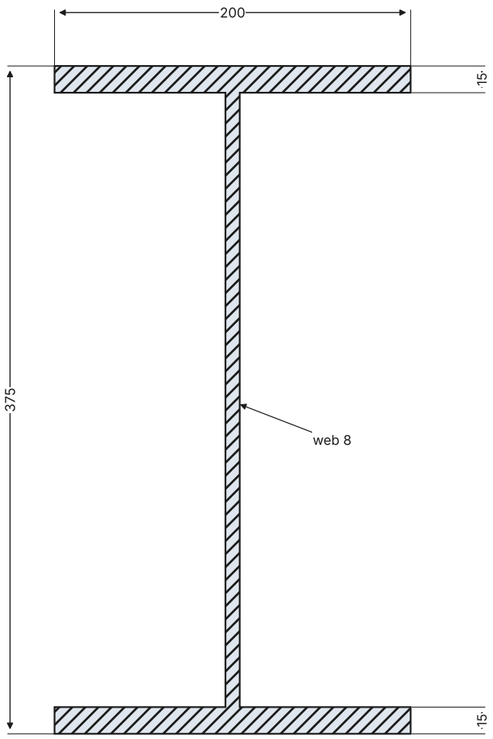
Notes
The section plane runs along the main-rib centreline; the rib web is cut (hatched); the panes either side are hidden by the web in that plane.
Frame: 140 deep at the seam; at the vertex the first 100 mm of each frame is the solid corner block (z 0 to 140). Its extent along the rib is not specified: drawn to x 115 (100 / cos 30 deg), undimensioned.
Main rib I 375 runs vertex to centroid, split at x 1,125 (splice not specified). At x 1,125 four secondary ribs (I 160) meet the main rib at 60 deg (joint not specified).
Seat lip z 93 to 103 and the L7 bite (20 on the seat, 10 clear of the web) are the frame values of spec sections 4 and 7; the rib seat is not specified and is left undimensioned.
Stresses at 1 atm: main rib 106 to 128 MPa; secondary 116 to 147; allowable 160.

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

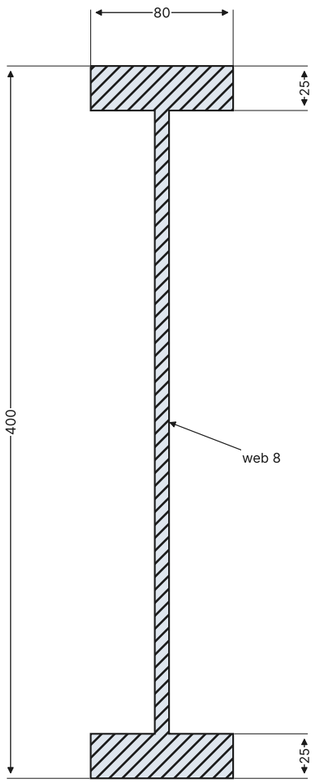
Not yet locked in TRUTH until Manos signs the Rev D revision. Rib seat for L7 and rib-to-block joint: not specified.

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 03: hexagon section A-A along a main rib, vertex to centroid	**DWG RD-03**
Material: 6061-T651 plate / 6061-T6511 extrusion (prototype); 2219-T851 (flight); L7 BOROFLOAT 33	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C	

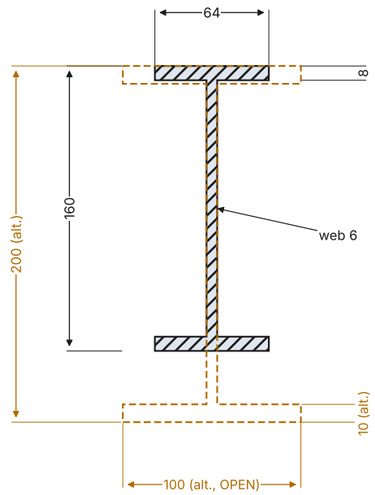
Rib sections, depth z into the cabin from the outer face (z 0 at the top); all drawn to one scale, NTS



Main rib, hexagon
I 375, flanges 200 × 15, web 8
6 per hexagon, 2,250 long
106 to 128 MPa at 1 atm

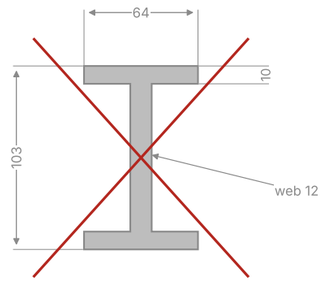


Main rib, pentagon
I 400, flanges 80 × 25, web 8
5 per pentagon, 1,914 long
114 to 140 MPa at 1 atm



Secondary rib
I 160, flanges 64 × 8, web 6
18 per hexagon, 1,125 long
116 to 147 MPa (hardpoint pinned)

Dashed: I 200, flanges 100 × 10,
web not specified; if the hardpoint
is moment-stiff (184 to 186 MPa
on I 160): OPEN until C3



Rev S rib: I 103, flanges 64 × 10,
web 12. SUPERSEDED: about
1,470 MPa in the Rev S run

Member	Section (depth, flanges, web)	A mm2	I mm4	Stress at 1 atm	Allowable	Flange b/t (<= 10)	Web h/t (<= 50)	Status
Main rib, hexagon	I 375, 200 × 15, web 8	8,760 (spec)	2.22e8 (spec)	106 to 128 MPa	160 MPa	6.4 (calc.)	43.1 (calc.)	FEA; 275 shallowest by hand, 250 fails
Main rib, pentagon	I 400, 80 × 25, web 8	6,800 (spec)	1.69e8 (calc.)	114 to 140 MPa	160 MPa	1.4 (calc.)	43.8 (calc.)	FEA
Secondary rib	I 160, 64 × 8, web 6	1,888 (spec)	7.41e6 (calc.)	116 to 147 MPa (pinned hardpoint)	160 MPa	3.6 (calc.)	24.0 (calc.)	FEA; OPEN 1 (C3)
Secondary rib, alternative	I 200, 100 × 10, web not specified	not specified	not specified	184 to 186 MPa on I 160 if moment-stiff	160 MPa	n/a	n/a	OPEN until C3
Rev S rib (superseded)	I 103, 64 × 10, web 12	2,276 (calc.)	3.35e6 (calc.)	about 1,470 MPa	160 MPa	2.6 (calc.)	6.9 (calc.)	superseded

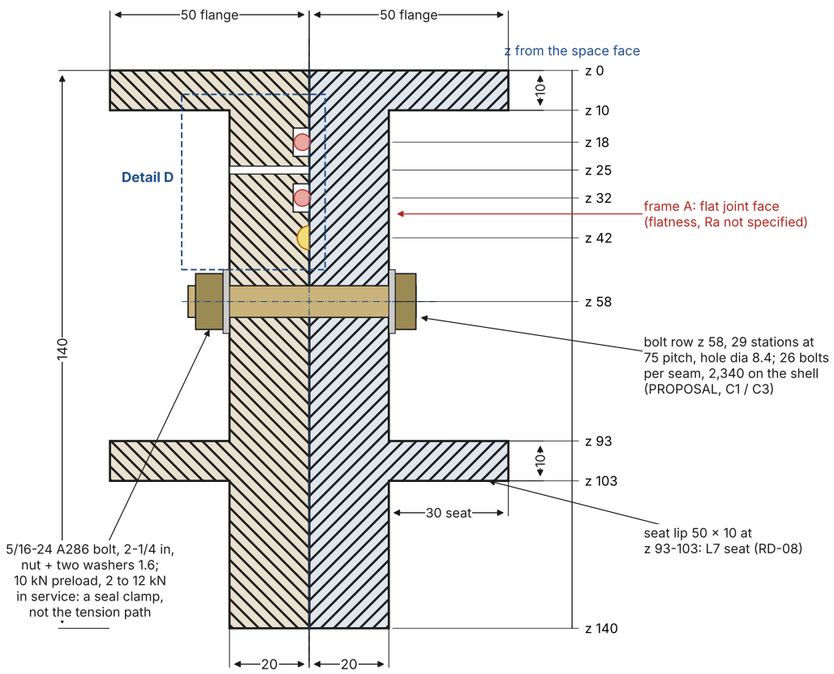
Allowable 160 MPa = 6061-T6 yield 240 / 1.5. Local buckling limits: flange outstand b/t <= 10, web h/t <= 50; all three Rev D sections pass both (spec section 3).
A and I marked (calc.) are computed from the section dimensions on this sheet (I = b h^3/12 - (b - tw)(h - 2 tf)^3/12); the spec gives A for all three and I for the hexagon rib only.
Depth is into the cabin from the outer face: the top flange is the outer (space) face at z 0. Rib seat for L7, rib splice at x 1,125 and rib end joints: not specified.

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision. Rib splice, end joints and L7 seat on the ribs: not specified.

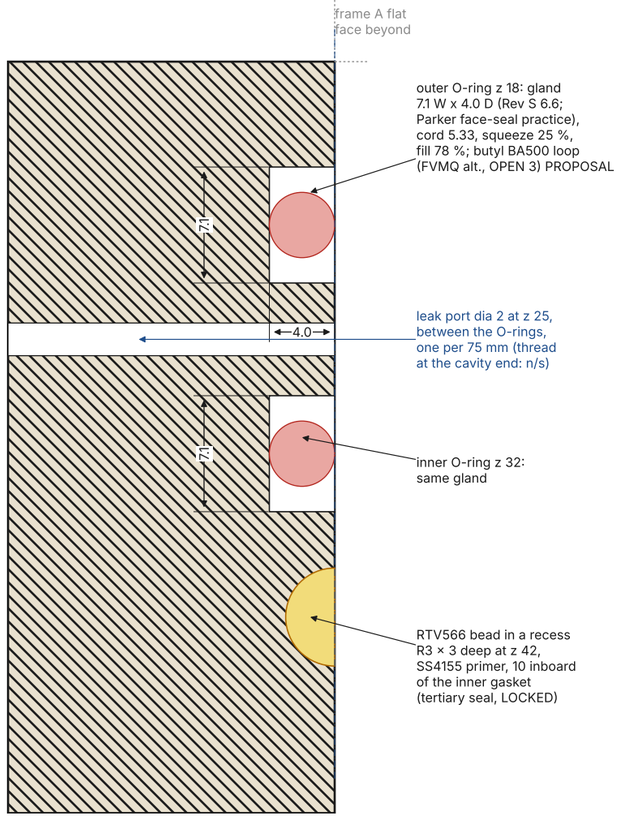
HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 04: rib sections, main hexagon, main pentagon, secondary	**DWG RD-04**
Material: 6061-T651 plate / 6061-T6511 extrusion (prototype); 2219-T851 (flight)	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C	

Section across a seam at a bolt station (y across, z down from the space face)

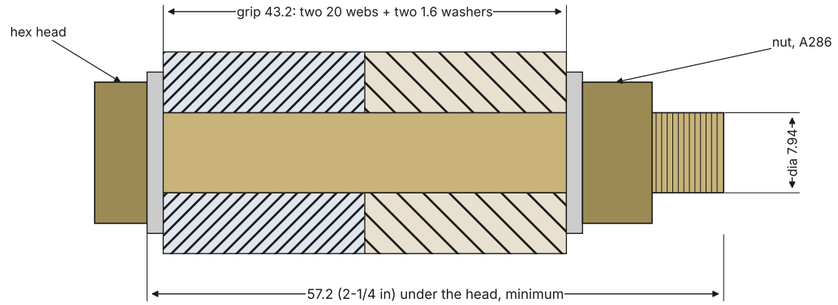


joint plane y 0 (radial): A flat, B grooved; two frames per seam

Detail D: frame B seal zone, z 8 to 50 (enlarged)

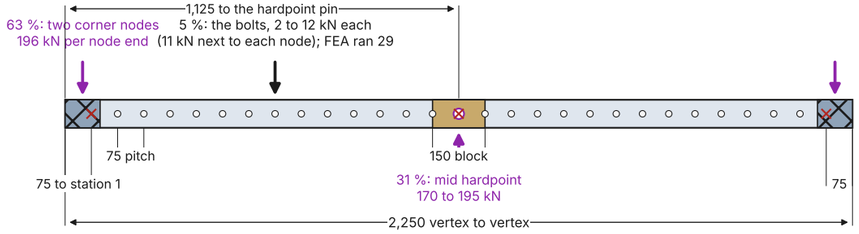


Seam bolt: 5/16-24 UNJF x 2-1/4 in, A286, nut and two washers



Designation OPEN 4 (NAS9306C-06 is a blind-rivet standard, 3/16 size; candidate NAS6605-type): 1 Nov memo

Load path along one seam (x 0 to 2,250): the 623 kN seam tension at 1 atm, FEA split



Bolt stations: 29 at 75 pitch, x 75 to 2,175 (30 collides at pentagon corners). Stations 1 and 29 (x 75, 2,175) fall inside the 100 mm corner blocks and are replaced by the vertex ties (RD-06); station 15 (x 1,125) is taken by the hardpoint pin (RD-07): 26 seal-clamp bolts per seam, 2,340 on the shell. Crossed stations above are the three replaced ones. PROPOSAL, confirmed in C1 and C3.

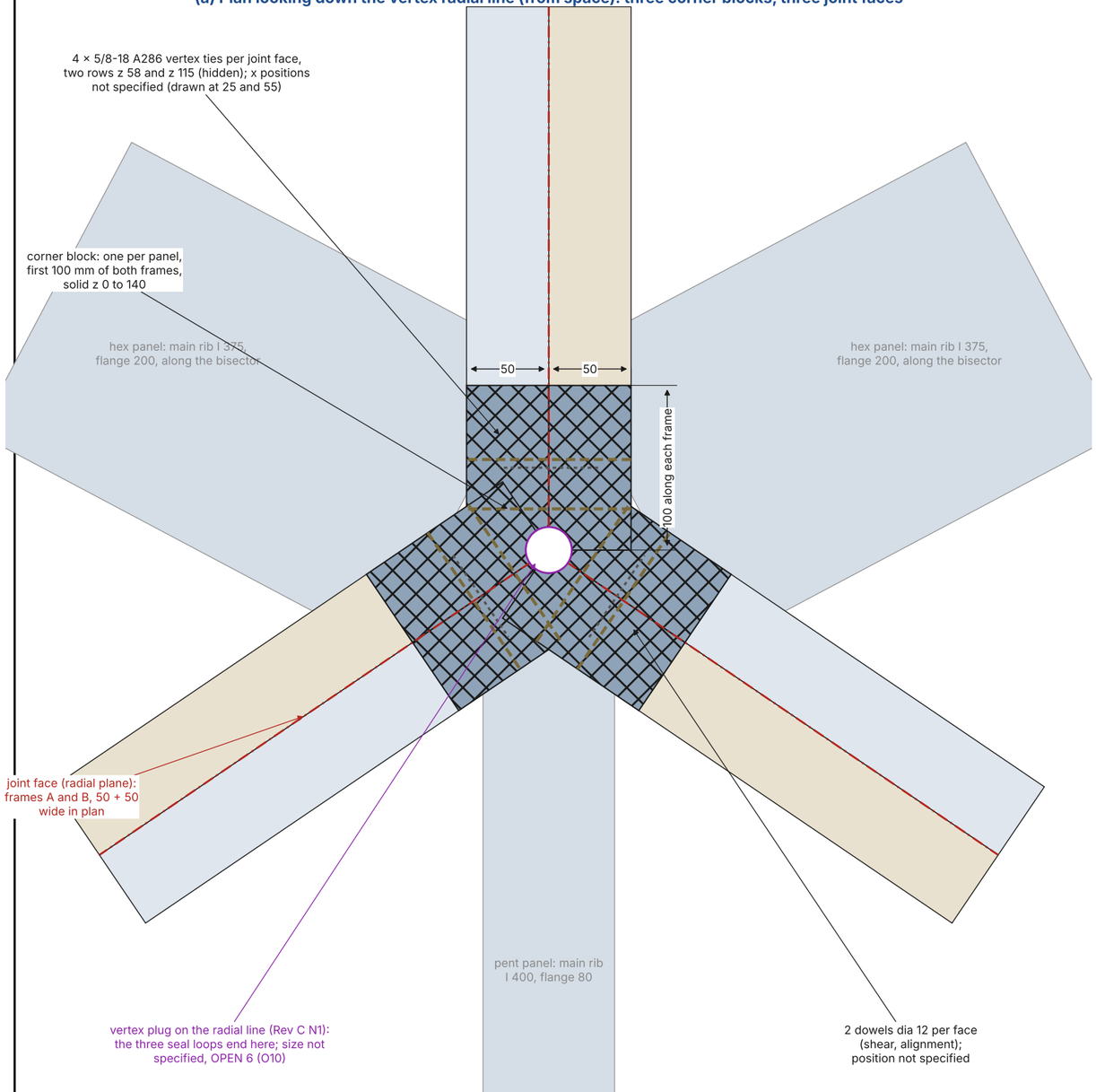
Seam tension 276.7 kN/m (pR/2 bound 282.5); 623 kN per 2.25 m seam. Frame stress 94 to 118 MPa; frame bow between corners 1.4 to 2.0 mm vs TRUTH 1 mm sealing limit: OPEN 5 (keep 140 deep and test in T1 / C2; a 300 deep frame still bows 1.07).
Rev S put the whole seam tension on the bolts (20.7 kN each): the slice pries open at 0.3 atm and exceeds the 17 kN bolt limit at 0.66 atm (D1).
Thermal -40 to +120 C (D6): bolt 8.0 to 13.3 kN, squeeze 24.2 to 26.2 %; the cold limit is the elastomer's (FVMQ -73 C, butyl -59 C).
Not specified in Rev D: joint-face flatness and finish, groove root / edge radii, port thread, where the seal loops end at the vertex plug (OPEN 6).

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision. Gland 7.1 × 4.0 is PROPOSAL (Rev S 6.6).

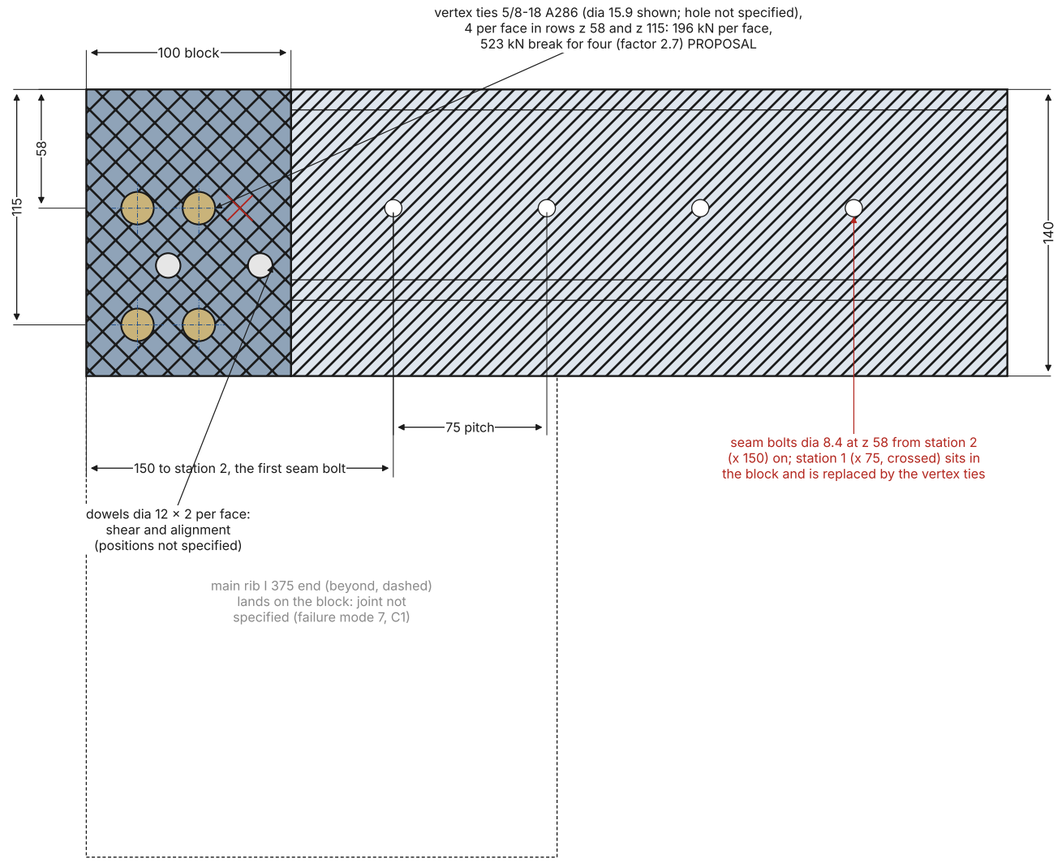
HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 05: seam joint section, frames A and B, bolt, seals, load path	**DWG RD-05**
Material: Frames 6061-T651; bolt 5/16-24 UNJF A286 AMS 5737; seals butyl BA500 (FVMQ alt.); RTV566	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C. A286: solution 980 C, age 720 C 16 h	

(a) Plan looking down the vertex radial line (from space): three corner blocks, three joint faces



Face angles are true 120 / 120 / 108 deg (sum 348: the 12 deg defect is the shell curvature); drawn opened to 124 / 124 / 112 deg so the plan closes.

(b) Section along the hex-hex seam through frame A (x from the vertex, z down): block, frame beyond, ties



Loads and status (spec section 5)
Each panel corner is one machined 6061-T651 block that includes the first 100 mm of both frames meeting there (the FEA node: the first 100 mm of all six frames at a vertex, solid). 5.8 kg per vertex, 350 kg on the shell (three blocks per vertex, one per panel).
Loads per panel corner at 1 atm: hexagon 380 to 430 kN with 50 to 65 kN m; pentagon 379 kN with 71 kN m.
Node stress 130 MPa (limit 160). Stiffness 4e6 to 5e6 N/mm, 4e10 to 1e11 N mm/rad.
Rev S corner (frames meeting on the miter, open) was 140 to 480x too soft in translation and 4 to 9x in rotation: superseded. Vertex tie and dowel sizes: C1 on 16 Oct (OPEN 7). Block outline beyond the two legs, fillets, tie hole size and the main-rib landing: not specified.

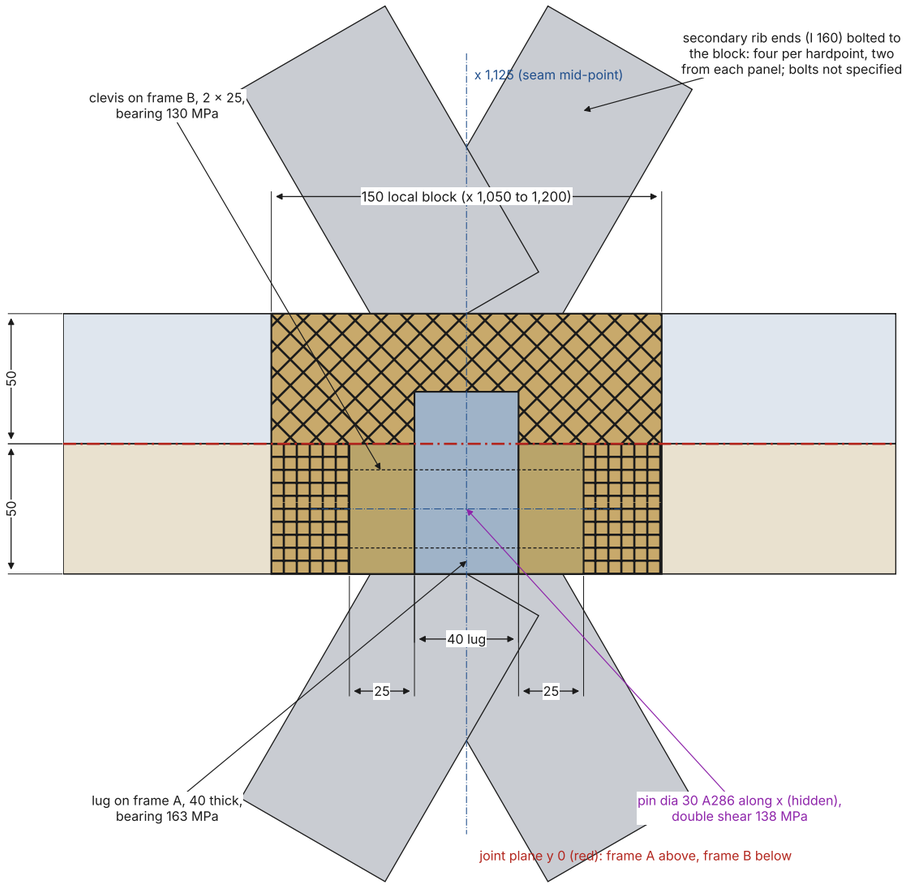
PROPOSAL: sized in C1, 16 Oct

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

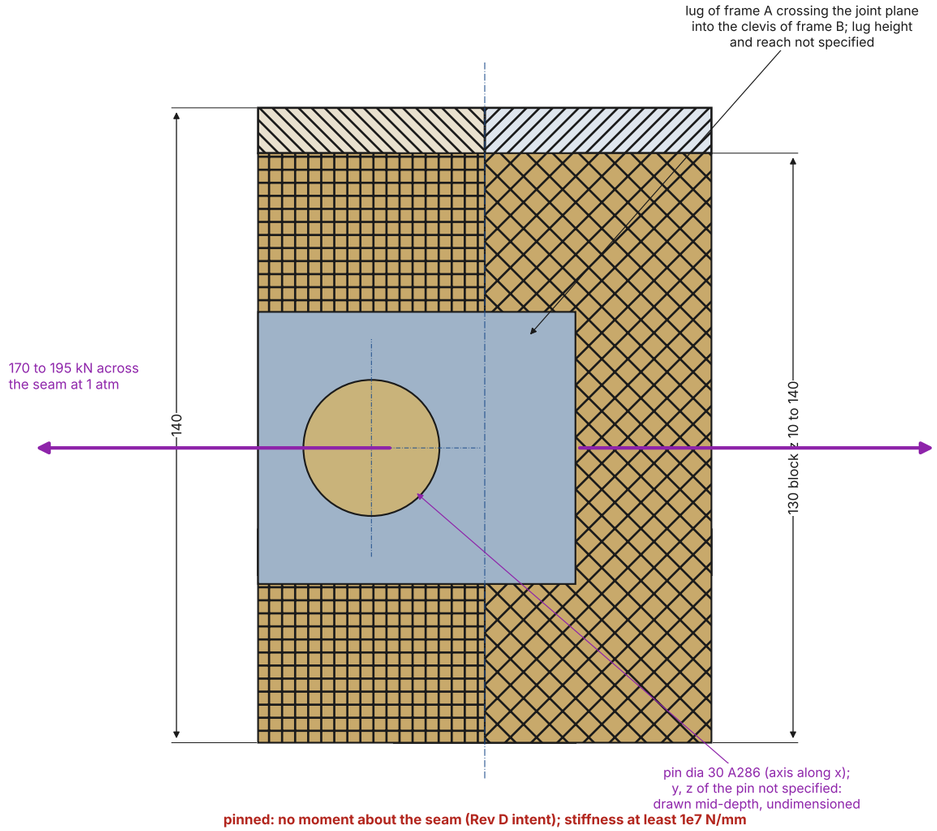
Not yet locked in TRUTH until Manos signs the Rev D revision. Tie and dowel positions along the face: not specified (C1); drawn clear of station 1 (x 75).

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 06: corner node, plan on the vertex and section along a seam (PROPOSAL)	**DWG RD-06**
Material: Blocks 6061-T651 machined; vertex ties 5/8-18 A286; dowels dia 12 (material not specified)	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C. A286: solution 980 C, age 720 C 16 h	

(a) Plan at the seam mid-point, viewed from the cabin (z 140 face): x along the seam, y across



(b) Section at x 1,125 through the pin (y across, z down): block z 10 to 140 in both frames



pinned: no moment about the seam (Rev D intent); stiffness at least 1e7 N/mm

Spec section 6
At the mid-point of every seam (90 on the shell), where four secondary ribs land (two from each panel). The lug on frame A (40 thick) sits between a clevis on frame B (2 x 25), overlapping across the joint plane; the pin dia 30 A286 runs along the seam (x) so the joint carries 170 to 195 kN across the seam and no moment about it. Pin double shear 138 MPa; lug bearing 163 MPa, clevis 130 MPa (A286 and 6061-T651 bearing allowables cover these; sized in C3). Local block 150 long (x) in both frames, z 10 to 140, secondary rib ends bolted to it. Stiffness at least 1e7 N/mm. If the built detail is moment-stiff the secondary rib becomes I 200 (RD-04, OPEN 1).
Seam bolt station 15 (x 1,125) is taken by the pin: the seam has 26 seal-clamp bolts at the other stations (RD-05). PROPOSAL, C3.

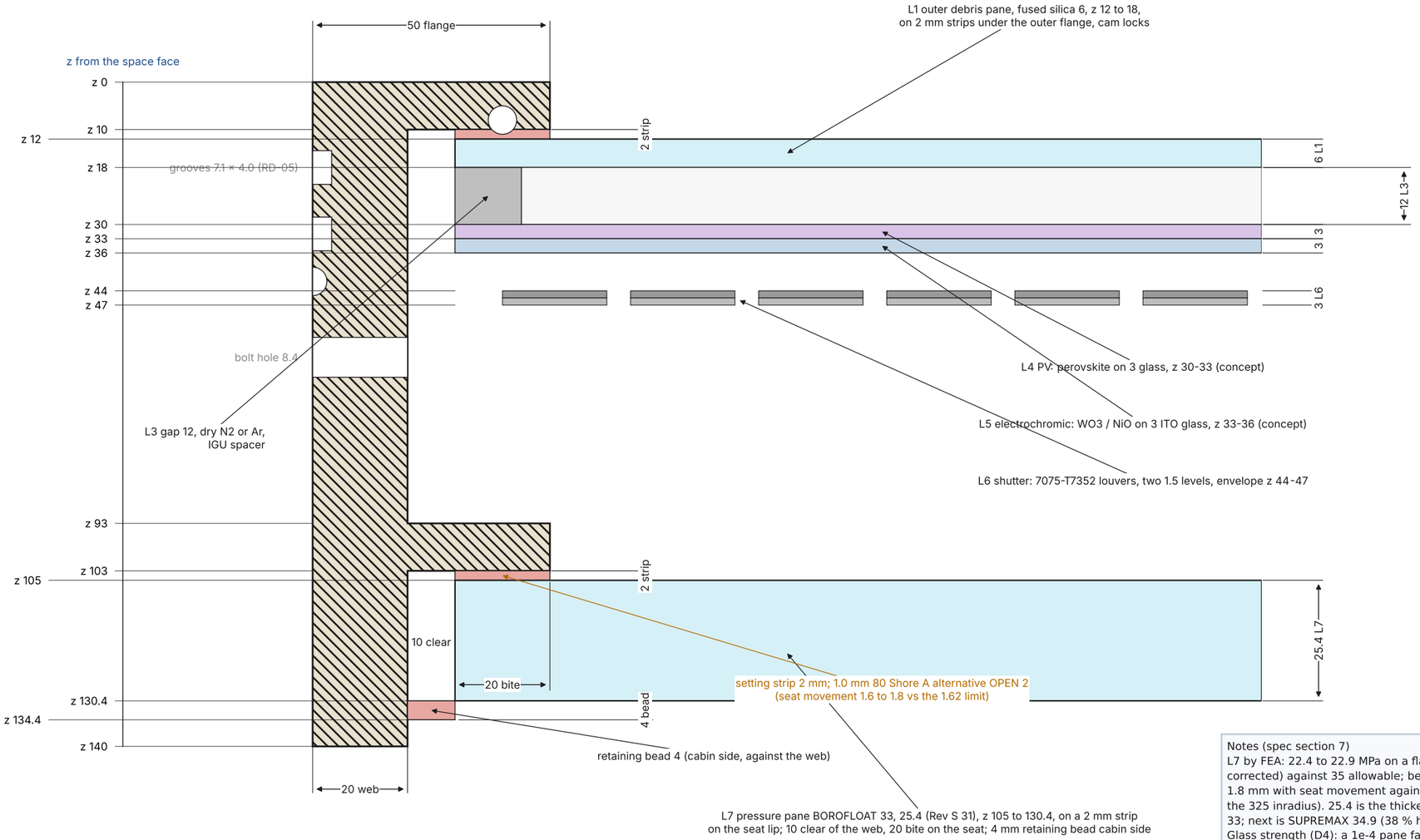
PROPOSAL: sized in C3, 16 Oct

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision. Pin position (y, z), lug height, block y-extent: not specified (C3).

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 07: seam mid hardpoint at x 1,125, plan and section (PROPOSAL)	**DWG RD-07**
Material: Blocks, lug and clevis 6061-T651; pin dia 30 A286	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C. A286: solution 980 C, age 720 C 16 h	

Pane edge at a frame: y from the joint plane (frame B, grooved face at y 0), z down from the space face; NTS, enlarged



Notes (spec section 7)
L7 by FEA: 22.4 to 22.9 MPa on a flat hinge (27 mesh-corrected) against 35 allowable; bending 1.03 mm; 1.6 to 1.8 mm with seat movement against the 1.62 limit (1/200 of the 325 inradius). 25.4 is the thickest standard BOROFLOAT 33; next is SUPREMAX 34.9 (38 % heavier): OPEN 2.
Glass strength (D4): a 1e-4 pane failure probability at 1 atm needs a characteristic disc strength of 412 / 176 / 94 / 57 MPa at m 5 / 7 / 10 / 15; T0 measures m on bars (RD-10).
Pane edge rule: 10 clear of the web, 20 bite on the seat.
Layers L1 to L6 unchanged from Rev S (L4, L5 concept stage).

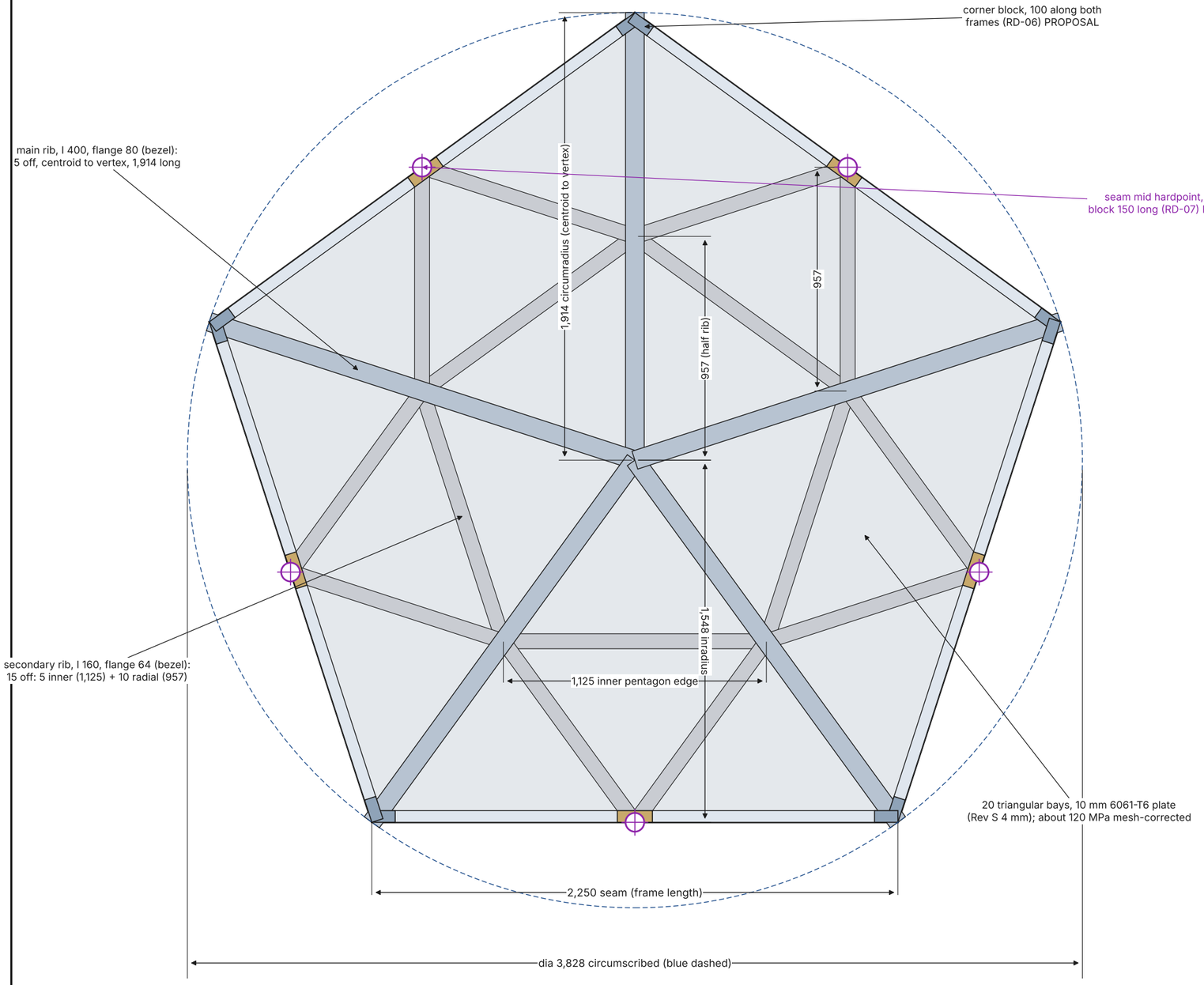
Not specified: lateral extent and edge support of L1 to L6 at the frame, cam-lock type, strip and bead materials, the pane edge finish class (T0 edge finishes), the rib-side seat.
Materials: L7 E 64 GPa, CTE 3.25; L1 E 72 GPa, CTE 0.5; frame 6061-T651 CTE 23.6 ppm/K.

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision. Lateral extent of L1 to L6 at the frame: not specified.

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 08: glazing stack and pane edge detail at the frame (frame B shown)	**DWG RD-08**
Material: L7 BOROFLOAT 33 fine annealed; L1 fused silica; frame 6061-T651; L6 7075-T7352; strips and bead: material not specified	
HT: BOROFLOAT 33 fine annealed by SCHOTT (not tempered, not chem. strengthened); frame T651 as RD-05	

Pentagon panel viewed from the cabin; one of 12 (opaque: 10 mm plate, services as TRUTH 8.4, 11, 12)



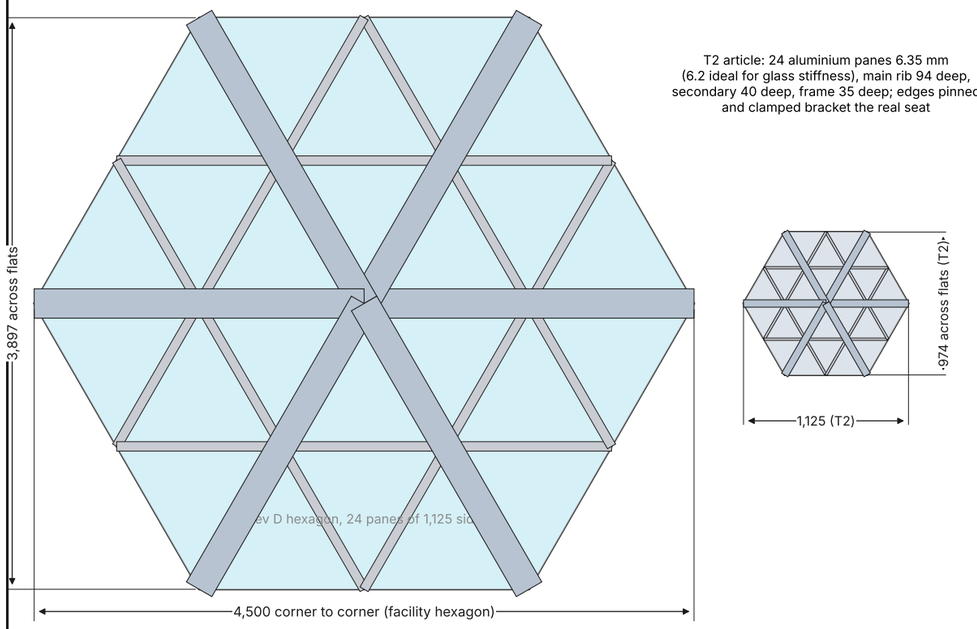
Notes (spec section 8)
5 main ribs (I 400, 80 x 25, web 8) centroid to vertex, 1,914 long.
15 secondary ribs I 160: 5 form the inner pentagon of 1,125 edge; 10 run from the rib mid-points to the two neighbouring seam mid-points, 957 long (two of each panel at every hardpoint).
20 triangular bays; secondary ribs 15.2 m, 77 kg.
Face plate 10 mm 6061-T6 (Rev S 4 mm), about 120 MPa; 235 kg.
Face load at 1 atm 883 kN; 8.710 m2.
Tolerance: a 0.2 % pentagon oversize is a 140 MPa self-stress; pentagon frames need the same machining tolerance as the hexagons (0.02 % evens the corner loads by 9 %). Frame section, seam, corner and hardpoint as RD-05 to RD-07 (a pentagon is always frame B: Rev S rule, unchanged).
Not specified: plate-to-rib attachment, plate edge at the frame, rib seat for the plate, rib splice, variant equipment.

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision. Plate-to-rib attachment and plate edge at the frame: not specified.

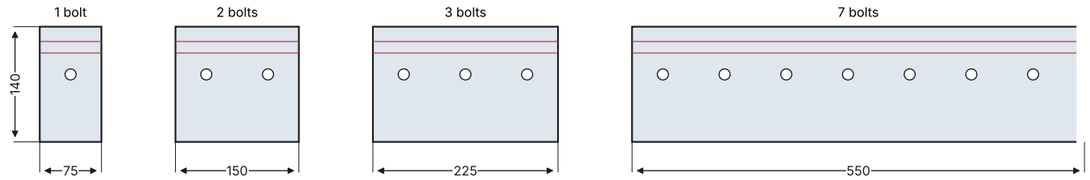
HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 09: pentagon panel plan, cabin side, ribs, 20 bays, 10 mm plate	**DWG RD-09**
Material: Frame, ribs 6061-T651 / T6511; face plate 10 mm 6061-T6	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C	

T2: 1:4 hexagon (1,125 corner to corner) beside the full panel (4,500), same drawing scale



T2 article: 24 aluminium panes 6.35 mm (6.2 ideal for glass stiffness), main rib 94 deep, secondary 40 deep, frame 35 deep; edges pinned and clamped bracket the real seat

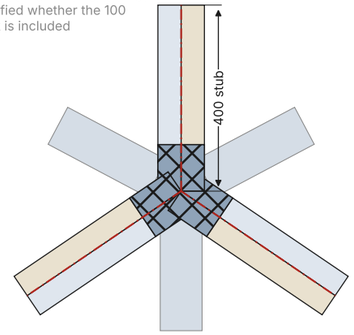
T1: seam slices 75 / 150 / 225 and the 550 mm article, full-size Rev D section (elevation on the joint face, x along the seam, z down)



75 / 150 / 225: pulled on the Instron 5882 after 18 Nov (D1 twin); 550: 7 bolts, not pulled (torque for 20 to 25 % squeeze; 1 atm between the seals holds 1 h). Bolt x positions in the slices: 37.5 + 75 k (rig pack Rev B).

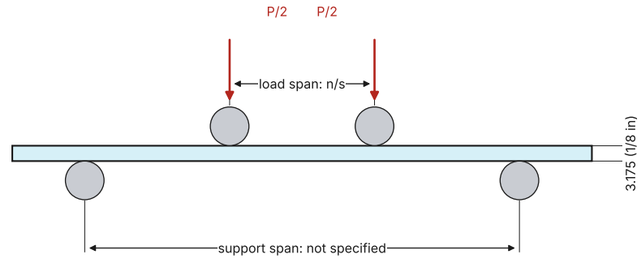
T3: one vertex, three corner blocks, six frame stubs 400 long, rib stubs, vertex ties (plan)

stub 400 from the vertex: not specified whether the 100 block is included



load applied along the seats, never by gripping stub ends; seam shear + dihedral moment need a three-actuator rig

T0: BOROFLOAT 33 bar 1 in x 1/8 in, four-point bend



width 25.4 (1 in) into the page; 30 bars per edge finish; alumina reference; ENGN 1440 spring 2027

Test	Article	Scale and what scales	What does not scale, and the guard	Success (pass rule, spec section 12)
T0 glass strength (ENGN 1440, spring 2027)	BOROFLOAT 33 bars 1 in x 1/8 in, four-point bend, 30 per edge finish; alumina reference	Strength does not scale geometrically: it is statistical (Weibull). Pane strength = bar strength x (bar effective area / pane effective area)^(1/m)	m is measured, not assumed; a second bar size checks the exponent between two sizes before it is trusted at 0.55 m2; edge finish classes separately	m and characteristic strength per finish; the predicted 1e-4 pane strength above the 27 MPa the pane sees (35 design); fractures start on the surface or edge as the finish class predicts
T1 seam slices (Instron 5882, after 18 Nov)	75 / 150 / 225 mm of the full-size seam, Rev D section	Full size; only the length is cut. Per-bolt response must be flat from 75 to 150 to 225 (the middle bolt of the 225 sees both neighbours, as every bolt in a seam does)	End dams, cut ends, grip legs: held by the D1 twin that models the slice and the 2,250 seam with the same code	At the Rev D service load (12 kN per bolt, 24 kN on the 150) inner gap under 0.266 mm and no leak; up the D1 curve within 10 % to 0.55 atm; bolt-station stiffness measured (D1 70 to 80 kN/mm) and fed to the shell model
T2 one panel at 1:4 (water rig, spring 2027)	Hexagon 1,125 corner to corner, 24 aluminium panes 6.35 mm, ribs 94 and 40 deep, 24 bays	Geometric similitude: at the same pressure a model with every length / 4 (thicknesses too) has the same stress and 1/4 the deflection; aluminium panes at (64/69)^(1/3) x 6.35 = 6.2 mm mimic the glass stiffness; edges pinned and clamped bracket the real seat	Setting strips and beads (rubber), bolt preload and friction, 2 mm webs (local buckling, machining), fillets, gravity, glass strength (never scales: T0). Guard: the panel model is built at both scales from one script; the items that do not scale are varied in the model to show they do not move the answer	Every gauge within 10 % at 1 atm at the matching edge condition; linear and repeatable over three cycles; the measured edge condition sits between pinned and clamped and is adopted in the full-scale model
T3 corner (2027)	One vertex: three corner blocks, six frame stubs 400 long, rib stubs, vertex ties	Full size. Load applied along the seats (never by gripping stub ends)	The seam shear and the dihedral moment need a three-actuator rig or a reacting fixture	Rosettes at the junction within 10 % of C1; node stiffness measured and fed to the shell model
T1-550 article	550 mm straight seam, 7 bolts	Full size	Not pulled	Torque for 20 to 25 % squeeze; 1 atm between the seals holds for 1 h with the agreed decay
1:10 shell (only if the cap model disagrees with T2)	PLA or aluminium shell, 1,115 across	Load sharing is geometric and scales; stresses do not	Material	Load distribution between panels, seams and corners matches the shell model

The rule for every test: the digital twin files its prediction for every gauge before the test; the test passes when the measurement lands within 10 % at the first hold point; a miss changes the model, never the data; once a twin is trusted at test scale it is run at full scale and that is the scaling.
Facility: 11,151 across, 2,250 seams, 4,500 hexagons; T2 is the same hexagon at 1,125; T1 and T3 are the real seam and corner at 1:1; T0 is the real glass, scaled by statistics.

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision.

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 10: test articles against the facility, T0, T1, T2, T3	**DWG RD-10**
Material: T2 aluminium panes and ribs (alloy not specified); T1 / T3 6061-T651 Rev D section; T0 BOROFLOAT 33 bars; rigs 4140 Q&T, A36, A2	
HT: Rigs: 4140 Q&T 28-32 HRC, A2 58-60 HRC (rig pack Rev B); frames T651 as RD-05	

Materials and heat treatment (spec section 10)

Part	Material	Heat treatment	Properties used
Frames, ribs, corner blocks, hardpoint blocks, pentagon plates (prototype)	6061-T651 plate / 6061-T6511 extrusion	Solution 530 C, water quench, stretch 1.5 to 3 %, age 160 C 18 h; never welded (HAZ loses 40 to 50 % of yield), never above 150 C	Yield 240, UTS 290, E 68.9 GPa, CTE 23.6 ppm/K, allowable 160
Same parts (flight)	2219-T851	Solution 535 C, quench, stretch, age 175 C 18 h	Yield 352, UTS 455
Seam bolts, vertex ties, hardpoint pins	A286	Solution 980 C, age 720 C 16 h	UTS 895 min (130 ksi), yield 655, CTE 16.5
Shutter (L6)	7075-T7352	Solution 480 C, quench, stress-relieved, two-step age (about 107 C then 163 to 177 C)	Yield about 435
L7 pressure pane	BOROFLOAT 33	Fine annealed by SCHOTT; not tempered, not chemically strengthened	E 64 GPa, CTE 3.25, allowable 35 (design), strength from T0
L1 outer debris pane	Fused silica	Annealed	E 72 GPa, CTE 0.5
Seals	Butyl BA500 or FVMQ	Post-cured by the maker	See RD-05 (spec section 9)
Test rigs	4140 Q&T 28 to 32 HRC (blocks, adapters), A36 (tub), A2 58 to 60 HRC (rollers)	As in the rig pack Rev B	

Mass (spec section 11; FEA sections, first order; proposals for nodes and hardpoints included)

Item	Per panel (kg)	Shell (kg)
Hexagon: frame 124, main ribs 319, secondary ribs 103, L7 745, L1 174, L4 88, L5 88, L6 111, hardware 50, node share 12, hardpoint share 24	1,838	36,760 (20)
Pentagon: frame 103, main ribs 176, secondary ribs 77 (15 ribs, 15.2 m), plate 235, hardware 30 (base, before variant equipment)	621	7,450 (12)
Corner nodes (in the panel figures)		350
Structure only (frames, ribs, L7, plates, nodes)		33.3 t (Rev S 25.4 t, which fails; 34.9 mm glass would be 38.5 t)
Facility (TRUTH 13.2 non-panel items unchanged: 2,480 kg stand-alone; pentagon variant equipment as 13.2)		about 50.9 t (Rev S 41.7 t)

Notes
Launch: Starship V3 (100 t conservative) keeps about 49 t margin; Falcon Heavy reusable (50 t) is no longer a single-launch option; Falcon Heavy expendable (63.8 t) remains.
The 25 t structure target of Rev S is withdrawn (OPEN 8). Mass comes down next from a transparency-led rib study (narrower, deeper ribs with a thicker web recover clear area at similar stress; curved glass is the untested hypothesis that could remove most ribs).

Check: hexagon items listed sum to 1,838 (124 + 319 + 103 + 745 + 174 + 88 + 88 + 111 + 50 + 12 + 24); pentagon items sum to 621 (103 + 176 + 77 + 235 + 30).
Seam bolts: 26 per seam, 2,340 on the shell (29 stations at 75 pitch less the two corner-block stations and the hardpoint station), PROPOSAL (RD-05).
Arithmetic on sheet RD-04: hexagon main rib 6 x 2.25 m x 8,760 mm2 x 2.70 g/cm3 = 319 kg (matches); secondary 18 x 1.125 m x 1,888 mm2 = 103 kg (matches); pentagon secondary 77 kg = 15.2 m of I 160 (5 x 1,125 + 10 x 957), see RD-09. Density 2.70 is a drafting check value, not a spec number.

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision.

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 11: materials, heat treatment and mass	**DWG RD-11**
Material: as tabled (spec sections 10 and 11)	
HT: as tabled	

Open decisions, with the Rev D default drawn on the sheets named

#	Decision (spec section 13)	Rev D default	Alternative	Settled by	Sheets
1	Hardpoint pinned or moment-carrying: sets the secondary rib	Pinned; secondary rib I 160 (64 × 8, web 6)	Moment-stiff; secondary rib I 200 (100 × 10)	C3 on 16 Oct	RD-04, RD-07
2	L7 thickness and seat	25.4 BOROFLOAT 33 with a stiffer setting strip (1.0 mm, 80 Shore A)	28 to 34.9 (SUPREMAX 34.9, 38 % heavier)	After C4 and T0 (and T2 for the seat)	RD-03, RD-08
3	Seal cord	Butyl BA500 closed loop in 7.1 × 4.0 glands	FVMQ AMS-R-25988 (TRUTH Rev S baseline)	Before 8 Nov (drawings)	RD-05
4	Seam bolt designation	5/16-24 UNJF A286 hex bolt, 2-1/4 in (candidate NAS6605-type)	'NAS9306C-06' is a blind-rivet standard, 3/16 size: wrong	1 Nov parts memo	RD-05
5	Frame bow limit 1 mm (TRUTH) against 1.4 to 2.0 predicted	Keep 140 deep and test	Deepen the frame (300 deep still bows 1.07)	After T1 and C2	RD-05
6	Corner seal at the vertex plug (O10)	Three seal loops end at a vertex plug on the radial line (Rev C N1)	Not stated	Not dated (O10)	RD-06
7	Vertex tie and hardpoint pin sizes	4 × 5/8-18 A286 per joint face, two rows; pin dia 30 A286	Sized up or down by the submodels	C1 and C3 on 16 Oct	RD-06, RD-07
8	Mass	Rev D structure 33.3 t; 25 t target withdrawn	Transparency-led rib study is the next pass; curved glass hypothesis	Next design pass	RD-11

What each test settles

Test	When	Article	What it settles	Pass rule (spec section 12)
T0 glass strength	ENGN 1440, spring 2027	BOROFLOAT 33 bars 1 in x 1/8 in, four-point bend, 30 per edge finish; alumina reference	Weibull m and characteristic strength per edge finish; with D4 rerun for bars (16 Oct) the 1e-4 pane strength; feeds decision 2 (L7 thickness)	Predicted 1e-4 pane strength above the 27 MPa the pane sees (35 design); fractures start where the finish class predicts
T1 seam slices	Instron 5882, after 18 Nov	75 / 150 / 225 mm Rev D seam	Bolt-station stiffness (D1 70 to 80 kN/mm) for the shell model; seal gap and leak at service load; decision 5 (frame bow) with C2; decision 3 (cord) by the 550 article	Inner gap under 0.266 mm and no leak at 12 kN per bolt; within 10 % of D1 to 0.55 atm
T1-550 article	Before T1	550 mm straight seam, 7 bolts	Bolt torque for 20 to 25 % squeeze (the preload used on the shell); seal choice (decision 3)	1 atm between the seals holds 1 h with the agreed decay
T2 one panel at 1:4	Water rig, spring 2027	Hexagon 1,125 corner to corner, 24 aluminium panes 6.35, ribs 94 / 40, frame 35	Pane and rib stresses and deflections of the 24-pane panel; the real edge condition between pinned and clamped (decision 2, seat); whether a 1:10 shell is needed	Every gauge within 10 % at 1 atm at the matching edge condition; linear and repeatable over three cycles
T3 corner	2027	One vertex: three blocks, six 400 stubs, rib stubs, vertex ties	Node stiffness and junction strains against C1 (decisions 6 and 7: vertex ties, corner seal)	Rosettes within 10 % of C1; node stiffness fed to the shell model
C1 / C2 / C3 / C4 (Ansys submodels, 16 Oct)	16 Oct	Corner node / frame and seam / hardpoint / L7 seat	Sizes for the PROPOSAL items (decisions 1, 5, 7) and the seat stiffness (decision 2); the predictions the tests are checked against	Within 10 % at the first hold point of each test, or the model changes
1:10 shell	Only if the cap model disagrees with T2	PLA or aluminium shell, 1,115 across	Load sharing between panels, seams and corners	Matches the shell model

The rule for every test: the digital twin files its prediction for every gauge before the test; the test passes when the measurement lands within 10 % at the first hold point; a miss changes the model, never the data; once a twin is trusted at test scale it is run at full scale and that is the scaling.
Rev D is not locked in TRUTH until Manos signs the Rev D revision; the sheets carry the FEA sizes and the PROPOSAL details so that the November tests test the design we mean to build.

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision.

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 12: open decisions and what each test settles	**DWG RD-12**
Material: n/a (no part on this sheet)	
HT: n/a	